Third Terminal Examination - 2079

Class : XI | **Subject: Physics (1011)** | **Full Marks: 75**

Time: 3 hrs. | **SET 'A'** | **Pass Marks: 30**

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group A

Rewrite the best alternative of each question in your answer sheet. [11 × 1 = 11]

1. A meter rule is used to measure the length of a piece of string in a certain experiment. It is found to be 20 cm long to the nearest millimeter. How should this result be recorded in a table of results?

   a. 0.2000m   b. 0.200m
   c. 0.20m   d. 0.2m

2. Which of the following have same dimension?

   a. Angular momentum and linear momentum
   b. Work and power
   c. Work and Torque
   d. Torque and pressure

3. Whatever may be the direction of two forces of 6N and 2N acting on a body of mass 2kg simultaneously, then minimum acceleration produced in the body in ms^{-2} is

   a. 1   b. 2   c. 3   d. 4

4. A man of mass 80kg is standing on a bathroom scale inside lift that is moving downward at constant retardation of 5 ms^{-2}. The reading of scale is

   a. zero   b. 400N
   c. 800N   d. 1200N

5. A ball of mass m_1 and another ball of mass m_2 are dropped from equal height. If the time taken by balls are t_1 and t_2 respectively to reach ground, then

a. $t_1 = t_2$

b. $\frac{t_1}{t_2} = \frac{m_2}{m_1}$

c. $\frac{t_1}{t_2} = \frac{m_1}{m_2}$

d. $\frac{t_1}{t_2} = \sqrt{\frac{m_2}{m_1}}$

6. Density of a liquid at 0°C is 13.6 g/c.c. Its density at 100°C is ($\gamma = 5.3 \times 10^{-4}/°C$)

a. 13.6 g/c.c.

b. 12.91 g/c.c.

c. 14.32 g/c.c.

d. 13.02 g/c.c.

7. An object is placed at 20 cm from a convex mirror of focal length 10 cm. The image formed by the mirror is:

a. Real and 20 cm from the mirror.

b. Virtual and 20 cm from the mirror.

c. Real and 20/3 cm from the mirror.

d. Virtual and 20/3 cm from the mirror.

8. When light passes through glass slab

a. wavelength decreases

b. wavelength increases

c. velocity increases

d. frequency decreases

9. A prism of refracting angle 60° produces a minimum deviation of 30°. Then the angle of incidence is

a. 30°

b. 60°

c. 45°

d. 90°

10. 'n' small drops of same size are charged to V volt each. If they coalesce to form a single large drop, then its potential will be

a. V n

b. $V\ n^{-1}$

c. $V\ n^{1/3}$

d. $V\ n^{2/3}$

11. NC^{-1} has the same unit as

a. Volt – meter

b. Volt/meter

c. Farad – m

d. Farad/m

Group – B

Give short answer to the following questions. **[8 × 5 = 40]**

12. a. Suppose you are in a trip to Mustang with your family in a Highway on a car. Suddenly rain starts to fall and your small sister shows you diagonal streaks of rain in the side windows of car. How and why, they are formed even though the water is falling vertically? [2]

    b. A disoriented physics professor drives 3.25 km north, then 4.75 km west and then 1.50 km south. Find the magnitude and direction of the resultant displacement using the method of resolution. [3]

13. a. State the law of conservation of momentum. [2]

    b. A body of mass 1kg is dropped from height of 20m. On striking the ground it comes to rest in 0.01s. Calculate the average force exerted by ground on body. [2]

    c. After landing on ground, body's momentum becomes zero. Explain how the law of conservation holds here. [1]

14. A light rope is attached to a block with mass 4Kg that rests on a frictionless, horizontal surface. The horizontal rope passes over a frictionless pulley and a block with mass m is suspended from the other end. When the blocks are released the tension in the rope is 10N.

    a. Draw free body diagram of the problem. [1]

    b. Calculate the acceleration of either block. [2]

    c. Find the mass 'm' of the hanging block. [2]

15. a. A newspaper article about the weather states that " the temperature of body measures how much heat the body contains. " Is this description correct? Why or why not? [1]

b. A substance takes 3 minutes in cooling from 50°C to 45°C and takes 5 minutes in cooling from 45°C to 40°C. What is the temperature of the surroundings? [3]

c. Why gas thermometers are more sensitive than mercury thermometers? [1]

16. a. The length of an iron rod is measured by a brass scale. When both of them are at 10°C, the measured length is 50cm.

i. What is the length of the rod at 40°C when measured by the brass scale at 10°C.? [1]

ii. What is the length of the rod at 40°C when measured by the brass scale at 40°C?

(α for brass = 24×10^{-6} /°C, α for iron = 16×10^{-6}/°C) [2]

b. "Food cooked in pressure cooker is well cooked than in open pot." Why? [2]

OR

a. Does it really help to run hot water over a tight metal lid on a glass jar before trying to open it? Explain your answer. [2]

b. An underground tank with a capacity of 1700 litre (1.70 m^3) is filled with ethanol that has an initial temperature of 19.0°C. After the ethanol has cooled off to the temperature of the tank and ground which is at 10.0°C, how much air space will there be above the ethanol in the tank? (Assume that the volume of the tank doesn't change). [Given: γ for ethanol = 75×10^{-5}/°C] [3]

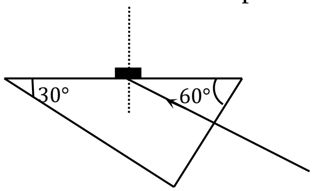
17. a. "A concave mirror is called converging mirror. Why? [1]

b. State one daily application of convex Mirror. [1]

c. A ray of light is incident at 60° in air on an air-glass plane surface. Find the angle of refraction in the glass (μ for glass = 1.5). [3]

18. a. Find the condition for no emergence of light from a prism. [2]

b. Light is incident normally on the shortest face of a 30° – 60° – 90° prism. A drop of liquid is placed on the hypotenuse of the prism. If the index of the prism is 1.50, find the maximum refractive index the liquid may have if the light is to be totally reflected. [3]



19. a. State Gauss law. [1]

b. Apply Gauss law to calculate the electric field at a point outside the hollow charged sphere. [2]

c. Sketch the graph to show the variation of electric field intensity with distance (r) from centre of hollow charged sphere. [2]

OR

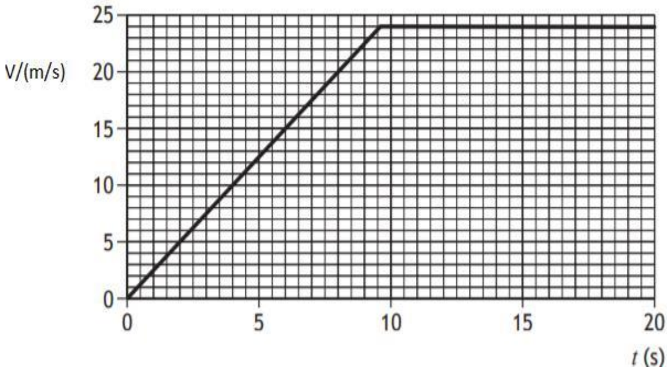
a. Five balls, numbered 1 to 5 are suspended using separate threads. Pairs (1,2), (2,4), (4,1) show electrostatic attraction, while pairs (2,3) and (4,5) show repulsion. what is the nature of ball 1? Explain [2]

b. A charged oil drop of mass 5×10^{-15} kg remain stationary when situated between two horizontal metal plates 5 mm apart and p.d. of 500 V is applied to plates. Find the charge on oil drop. [3]

Group – C

Give long answer to the following questions. **[3 × 8 = 24]**

20. a. A box at rest is accelerated by a rope attached with a motor as shown in the Figure. The velocity-time graph given below shows the pattern of its motion for 20 s.



i. If the box is pulled with constant unbalanced force 10N. Show that the initial acceleration of the box is 2.5 ms^{-2}, and calculate its mass. [2]

ii. After 2.0 second the box is being pulled by a constant force 12 N. Determine the size of frictional forces acting on the box at this time. [2]

iii. Determine the distance of the box travels along the ground at 8.0s. [2]

b. Show that Newton's third law of motion is contained in Newton's second law of motion. [2]

OR

a. What is friction? State the laws of friction. [2]

b. A block slides down from rest from the top of inclined plane of inclination 30°. The coefficient of sliding friction between block and inclined plane is 0.3. calculate

i. acceleration of block. [2]

ii. time taken by block to reach bottom of inclined plane if plane is 5m long. For above problem, free-body diagram is compulsory. [1]

c. The equation of plane progressive wave is represented as

$y = A\sin(\omega t - kx)$, where 't' is time and 'x' is distance. Obtain the dimensions of 'ω' and 'k'. [2]

d. If $\vec{A}.\vec{B}$=0, what is the angle between $\vec{A}$ and $\vec{B}$? [1]

21. a. In a report of an experiment to measure the specific latent heat of fusion of ice, a student wrote the following. "Ice at 0^oC was added to water in a calorimeter. When the ice had melted, measurements were taken. The specific latent heat of fusion of ice was then calculated."

i. Draw a labelled diagram of the apparatus used. [1]

ii. What measurements did the student take before adding the ice to the water? [1]

iii. What did the student do with the ice before adding it to the water? [1]

iv. How did the student find the mass of the ice? [1]

v. Give one precaution that the student took to get an accurate result. [1]

b. What is the result of mixing 10gm of ice at 0°C into 15gm of water at 20°C in a vessel of mass 100gm and specific heat 0.09 cal/gm °C? [Take latent heat of fusion of ice 80 cal/gm] [3]

22. a. Define potential gradient. [1]

b. Establish the relation of electric field intensity with potential gradient. [3]

c. Does electric field intensity in a region is zero imply potential is also zero? Explain with an example. [2]

d. Vehicles carrying inflammable fluid drag a chain along the ground. Why? [2]

- The End -